Shiftfly

A shift-routed interconnect for accelerator fabrics beyond pod scale

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Abstract

Google's TPU interconnect spent nine generations as a Cayley graph of an abelian group—a k-ary n-cube whose diameter grows as $\Theta(N^{1/n})$— before TPU 8i replaced it with *Boardfly*, a three-tier fully connected hierarchy of Dragonfly type with diameter 7 over 1,152 chips. We first audit every shipped TPU topology against the Moore bound and show that at TPU-class radix ($\Delta \approx 7$) a diameter-7 graph could in principle hold 391,910 chips, so Boardfly spends its diameter budget roughly two orders of magnitude below the ceiling. We then observe that Boardfly's global tier is *fully connected*, which does not scale: at 400,000 chips a group would need 12,499 optical links, so a further hierarchy level must be stacked and diameters add.

We propose **Shiftfly**: retain Boardfly's cheap local structure and replace the fully connected global tier with a generalized Kautz digraph realised on the existing optical circuit switch. At an identical budget of 40 optical ports per group, Shiftfly is a flat graph of guaranteed diameter $\lceil \log_d G \rceil$ over G groups, routes without tables by a shift register, and admits content addressing natively.

Our evaluation is deliberately two-sided. Shiftfly loses at one-pod scale— Boardfly achieves chip-level diameter 7 where Shiftfly needs 11—and wins beyond it, with lower diameter, lower mean distance and roughly 2.7× better spectral expansion at equal optical cost. The redundancy result is mostly negative: locality-aware *placement* accounts for the large majority of the achievable saving on shared content, and it helps both topologies, leaving Shiftfly's algebraic path-merging worth only a few percent. We report the metric trap that conceals this, state the failure modes, and argue that the honest case for Shiftfly is scaling and expansion, not deduplication.

Contents

1 Introduction

An accelerator interconnect is a graph, and the quantities that matter—worst case latency, average latency, bandwidth under adversarial traffic, behaviour under failure—are graph invariants: diameter, mean distance, bisection width and connectivity. This paper takes that view literally.

Section 2 catalogues every Google TPU topology as a graph. Section 3 measures each against the Moore bound, which yields the observation motivating the rest of the work. Section 4 formalises a second objective—the cost of delivering *the same* content to many endpoints—because agentic inference makes a large fraction of traffic semantically redundant. Section 5 gives the construction, Section 6 its properties with proofs, Section 7 the measurements, and Section 8 the ways it is wrong.

The contribution is not the graph family. Kautz digraphs have been known to be near-optimal for the directed degree–diameter problem since 1968, and generalized de Bruijn and Imase–Itoh digraphs have provided arbitrary-order versions since 1980–81. The contribution is the argument that this family is the right global tier for an accelerator fabric that already owns a reconfigurable optical switch, together with an honest measurement of what that buys and what it does not.

2 TPU topologies as graphs

Definition 1 (k-ary n-cube)**.** Let $k_1, \dots, k_n \geq 2$. The torus $T(k_1, \dots, k_n)$ is the Cayley graph of $\mathbb{Z}_{k_1} \times \cdots \times \mathbb{Z}_{k_n}$ with generating set $\{\pm e_1, \dots, \pm e_n\}$.

Proposition 2. *$T(k_1, \dots, k_n)$ has $N = \prod_i k_i$ vertices, is $2n$-regular when all $k_i > 2$, and has*

$$\operatorname{diam} T = \sum_{i=1}^{n} \lfloor k_i/2 \rfloor .$$

In particular a cubic torus has $\operatorname{diam} T = \Theta\big(N^{1/n}\big)$.

Proof. Distance in a Cayley graph of an abelian group with this generating set is the sum of per-coordinate cyclic distances, and the cyclic distance in $\mathbb{Z}_k$ is maximised at $\lfloor k/2 \rfloor$; the maximum is attained simultaneously in every coordinate. □

The exponent $1/n$ with $n = 3$ is the worst scaling in common use, and it was the right choice while the dominant traffic was nearest-neighbour: dense training stencils and ring all-reduce touch only adjacent chips, so diameter never appears on the critical path. Mixture-of-experts routing broke the assumption. All-to-all traffic pays the diameter on every token.

Table 1 collects the generations. Diameters for torus generations are analytic; pod shapes for v5p, v7 and v8t are inferred from published chip counts and are marked accordingly.

Boardfly. Google describes it as "a building block of four fully connected chips into a fully connected group of eight boards, with 36 of such groups fully connected into a TPU 8i pod". Thus a building block holds 4 chips, a group holds 8 building blocks, i.e. 32 chips, and a pod holds 36 groups, i.e. 1,152 chips. Each building block exposes 16 external links of which 11 serve the group in copper, leaving 5 per building block and hence 40 optical ports per group. Within a pod the 36 groups are fully connected through the optical circuit switch, consuming 35 of those 40.

Generation	Chips	Δ	D	Structure
v2 / v3	256 / 1,024	4	16 / 32	Cayley graph of $\mathbb{Z}_k^2$
v4	4,096	6	24	$\mathbb{Z}_{16}^3$, 4^3 cubes, OCS between cubes
v5e / v6e	256	4	16	$\mathbb{Z}_{16}^2$
v5p	8,960	6	≈ 32	3D torus (shape inferred)
v7 Ironwood	9,216	6	≈ 32	3D torus; chiplets add a 4th logical axis
v8t	9,600	6	≈ 33	3D torus, Virgo scale-out fabric
v8i	**1,152**	**7**	**7**	**Boardfly**: 3-tier fully connected

Table 1: Google TPU interconnects as graphs.

Lemma 3 (Boardfly's diameter decomposition)**.** *If the optical link between two groups terminates on one designated chip of one designated building block on each side, the chip-level pod diameter is* 7*.*

Proof. Reaching the group's egress chip costs at most three hops: one inside the source building block, one across the copper link to the building block owning the optical port, one inside that building block. The optical hop is one. The destination side is symmetric. Hence $3 + 1 + 3 = 7$, and the bound is attained. □

We adopt Lemma 3's accounting throughout. Traversing g inter-group hops passes through $g + 1$ groups, so

$$\text{chip hops} \;=\; \iota\,(g+1) + g, \qquad \iota = 3, \tag{1}$$

which reproduces 7 at $g = 1$. Our chip-level simulation of a Boardfly pod measures diameter exactly 7, confirming the port model.

3 The Moore-bound audit

Theorem 4 (Moore bound)**.** *A graph of maximum degree $\Delta \geq 3$ and diameter D has*

$$N \;\leq\; 1 + \Delta \sum_{i=0}^{D-1} (\Delta - 1)^i \;=\; \frac{\Delta(\Delta-1)^D - 2}{\Delta - 2}.$$

Proof. Fix v. Breadth-first from v reaches at most Δ vertices at distance 1 and, since each subsequent vertex has at most $\Delta - 1$ unused incident edges, at most $\Delta(\Delta-1)^{i-1}$ at distance i. Every vertex lies within distance D. □

Equality requires a Moore graph, and these exist only for $D = 1$, for cycles, and for $D = 2$ with $\Delta \in \{3, 7, 57\}$. The bound is therefore a ceiling, and the informative quantity is the ratio to it.

At $\Delta = 7$:

$$\text{Moore}(7,4) = 1{,}814, \quad \text{Moore}(7,5) = 10{,}886, \quad \text{Moore}(7,6) = 65{,}318, \quad \text{Moore}(7,7) = 391{,}910.$$

Two consequences. First, Boardfly attains diameter 7 at 1,152 chips, whereas $\Delta = 7$ permits diameter 4 at that order—so even at its own scale the design is about 1.75× off. Second, and more importantly, $\Delta = 7$ admits diameter 7 at almost exactly 400,000 chips. Boardfly therefore spends, at pod scale, the diameter that would in principle serve a 400,000-chip machine.

Remark 5. This is not a claim that a 400,000-chip diameter-7 machine is buildable. Moore-optimal graphs of large diameter are not known, and the bound ignores bisection, cabling length and fault tolerance—all of which Boardfly is buying with its slack. It is a claim that the slack is unusually large and worth interrogating.

The structural obstruction. Boardfly's global tier is a complete graph K_{36} on groups, so a group needs $G-1$ optical ports to reach G groups. With $G = 12{,}500$ (i.e. 400,000 chips at 32 chips per group) this is 12,499 ports against 40 available. Full connectivity does not scale; another hierarchy level must be added, and by (1) each added level costs four chip hops. This is the weakness Shiftfly attacks.

4 Two objectives

Latency is the standard objective: minimise $\operatorname{diam} G$ and the mean distance $\bar{\mathrm{d}}(G)$. The second objective is specific to agentic inference, where one user query fans out into many concurrent rollouts that share a system prompt, tool definitions and retrieved context, and therefore request the *same* content.

Definition 6 (Multicast cost)**.** Let h be the home vertex of a content item and S the set of requesters. The *unicast cost* is $U(S,h) = \sum_{v\in S} \mathrm{d}(v,h)$. The *tree cost* $T(S,h)$ is the number of edges in the union of shortest paths from each $v \in S$ to h, i.e. what a fabric with in-network replication pays. The *multicast efficiency* is $\eta(S,h) = U(S,h)/T(S,h)$.

Remark 7 (A metric trap)**.** η must not be optimised alone. It is a ratio, and a topology is rewarded for placing requesters *far* from the home vertex provided their paths overlap. A network in which everything is one hop away scores $\eta = 1$—the worst possible value—while paying the least. Our measurements exhibit exactly this inversion: Boardfly attains higher η than Shiftfly while paying higher absolute T. The quantity to minimise is T; η is diagnostic only.

The design problem is the bi-objective

$$\min_{G:\ \Delta(G)\le\Delta_{\max}} \Big(\ \bar{\mathrm{d}}(G),\quad \mathbb{E}_{S,h}\big[T(S,h)\big]\ \Big), \tag{2}$$

and the two conflict. Expander graphs minimise $\bar{\mathrm{d}}$ but random routes share few edges; trees maximise sharing and have poor diameter and bisection.

5 The construction

Definition 8 (Kautz digraph)**.** For $d \ge 2$ and $D \ge 1$, $K(d,D)$ has as vertices the words $u_1u_2\cdots u_D$ over an alphabet of size $d+1$ with $u_i \ne u_{i+1}$, and an arc $u \to u_2\cdots u_D\alpha$ for every $\alpha \ne u_D$.

Definition 9 (Shiftfly)**.** Retain Boardfly's building block (4 chips, complete) and group (8 building blocks, complete; 32 chips). Replace the complete global tier by a Kautz digraph on the groups, realised as a fixed permutation installed on the optical circuit switch. When the group count is not of the form $(d+1)d^{D-1}$, use the Imase–Itoh digraph on $\mathbb{Z}_n$ with arcs $i \mapsto -di - r \bmod n$, $r = 1,\dots,d$, which exists for every n.

Remark 10 (Port accounting)**.** In $K(d,D)$ a vertex u is both a successor and a predecessor of v exactly when $u_1 = u_D$, and then in exactly one neighbour: solving $u_2\cdots u_D\alpha = \beta u_1\cdots u_{D-1}$ forces $\beta = u_2$, $u_1 = u_D$ and $\alpha = u_2$. There are $d(d+1)$ such vertices out of $(d+1)d^{D-1}$, a fraction d^{2-D}. Undirected degree is therefore $2d$, or $2d-1$ on that thin set, so **out-degree** d **costs** $2d$ **bidirectional ports**. Matching Boardfly's 40 optical ports per group means $d = 20$, not $d = 40$; comparing at $d = 40$ silently doubles Shiftfly's hardware and invalidates the result. We report only $d = 20$, and the simulator asserts the edge counts of the two designs agree to within 5%.

6 Properties

Theorem 11 (Order, degree, diameter). *$K(d, D)$ has $N = (d + 1)d^{D-1}$ vertices, out-degree and in-degree d, and diameter exactly D.*

Proof. There are $d + 1$ choices for u_1 and d for each subsequent symbol, giving the order. Out-degree is the number of $\alpha \neq u_D$, namely d. For the diameter, given u and v the word $u_{k+1} \cdots u_D v_1 \cdots v_k$ is a legal vertex whenever the junction symbols differ, and applying the arc map D times transforms u into v; hence $\mathrm{d}(u, v) \leq D$. It is exactly D for u, v sharing no admissible overlap. □

Against the directed Moore bound $N \leq (d^{D+1} - 1)/(d - 1)$, $K(d, D)$ is within a factor $1 + O(1/d)$: it is essentially optimal among digraphs.

Theorem 12 (Shift routing). *For $u, v \in K(d, D)$ the sequence $w^{(k)} = u_{k+1} \cdots u_D v_1 \cdots v_k$, $k = 0, \ldots, D$, is a walk from u to v of length at most D, computable from the labels alone.*

Proof. $w^{(0)} = u$ and $w^{(D)} = v$, and $w^{(k+1)}$ is obtained from $w^{(k)}$ by deleting the leading symbol and appending v_{k+1}, which is the arc map. □

Theorem 12 is the operational point: routing requires no tables, no routing protocol and no convergence, at any scale. The destination label is the route.

Theorem 13 (Merge criterion). *Let u, u' route to a common v under Theorem 12. Their walks occupy the same vertex at step k if and only if u and u' agree in their last $D - k$ symbols. Consequently the first merge occurs at*

$$k^*(u, u') \;=\; D - \mathrm{lcs}(u, u'),$$

where lcs *is the length of the longest common suffix.*

Proof. By Theorem 12, $w^{(k)}$ depends on u only through $u_{k+1} \cdots u_D$ and on v through a prefix common to both walks. Equality of $w^{(k)}$ and $w'^{(k)}$ is therefore equivalent to equality of those suffixes. □

Theorem 13 says redundant requests collapse *structurally*: once two requests for the same item meet, the second can be served from an in-network cache at the meeting point, with no coordination, no directory and no protocol. It is a statement about the routing algebra.

Proposition 14 (Random labels give almost no merging). *If labels are assigned uniformly at random, then $\Pr[\mathrm{lcs}(u, u') \geq j] = d^{-j}(1 + o(1))$, so $\mathbb{E}[\mathrm{lcs}] = \frac{1}{d-1}(1 + o(1))$ and $\mathbb{E}[k^*] = D - \frac{1}{d-1}(1 + o(1))$.*

Proof. Symbols are independent given the no-repeat constraint, and each successive suffix symbol agrees with probability $1/d$ up to the constraint correction. □

At $d = 20$ this is $\mathbb{E}[\mathrm{lcs}] \approx 0.053$: essentially no merging before the destination. **The merge property is not free.** It is purchased by choosing the labelling so that groups likely to want the same content are suffix-close—and that is the co-design step, an embedding of the observed sharing hypergraph into the suffix tree.

Theorem 15 (Arbitrary order). *The Imase–Itoh digraph on $\mathbb{Z}_n$ with arcs $i \mapsto -di - r \bmod n$, $r = 1, \ldots, d$, has out-degree d and diameter at most $\lceil \log_d n \rceil$ for every $n \geq 2$.*

Proof sketch. Iterating the arc map k times sends i to $(-d)^k i - \sum_j r_j (-d)^{k-j} \bmod n$. As the r_j range over $\{1, \ldots, d\}$ the reachable set after k steps has size $\min(d^k, n)$, and covers $\mathbb{Z}_n$ once $d^k \geq n$. □

This removes the quantisation that would otherwise strand capacity: a Kautz graph exists only at orders $(d+1)d^{D-1}$, and a real machine is whatever size it is.

Proposition 16 (Connectivity)**.** *$K(d, D)$ has vertex connectivity d, the maximum possible at that degree.*

Thin global tiers are the usual fragility of hierarchical fabrics; maximal connectivity is the relevant defence.

7 Evaluation

All measurements come from the accompanying simulator, which is dependency-free and reproduces every table and figure from one script. Distances are exact by breadth-first search where the instance permits and sampled otherwise, and sampled figures are labelled as such. Spectral gaps are power-iteration estimates. Both designs receive 40 optical ports per group and identical 32-chip group internals, so the comparison is at matched optical cost (Remark 10).

7.1 Model validation

The chip-level Boardfly pod model has 1,152 vertices, maximum degree 7, and **measured diameter exactly** 7, matching the published figure and Lemma 3.

7.2 Scaling

Figure 1 reports chip-level diameter against system size. **Boardfly wins at one-pod scale** (7 against Shiftfly's 11 at 1,152 chips) and loses beyond it, because the complete global tier must be re-hierarchised while the Kautz tier merely grows one symbol. Mean distance and spectral expansion follow the same pattern; at the scales tested Shiftfly's spectral gap is roughly 2.7× Boardfly's, which is unsurprising in hindsight: a hierarchy of near-cliques joined by few links is a poor expander by construction, whatever its diameter.

7.3 Scaling: the numbers

At matched optical cost, chip-level worst-case distance is:

Chips	Boardfly	Shiftfly
1,152	**7**	11
4,608	15	**11**
18,432	15	15
73,728	19	**15**
268,800	23	**15**
400,000	23	**19**
1,152,000	23	**19**

At the 400,000-chip target the reduction is $23 \to 19$, or 17%; the largest gap in the sweep is $23 \to 15$ at 268,800 chips, where Boardfly has just paid for another hierarchy level and Shiftfly has not yet needed another symbol. Both curves are step functions, so the advantage is not monotone.

7.4 Sharing

This is where the proposal is weakest, and we state it plainly. Writing T for tree hops per requester:

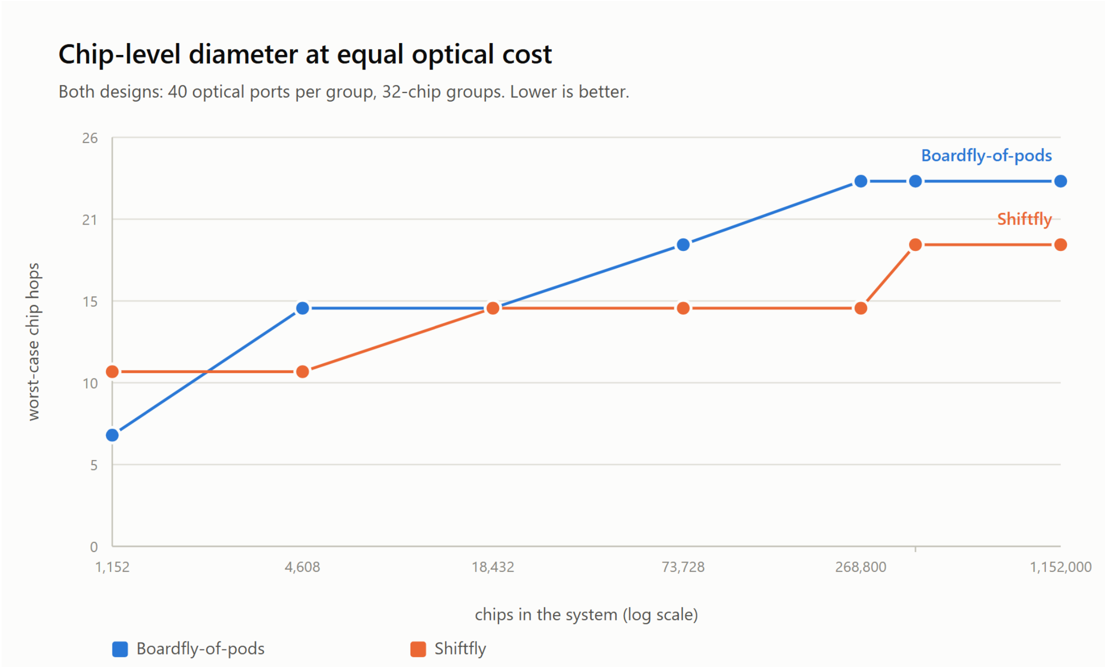

Figure 1: Chip-level worst-case distance at equal optical cost. Boardfly is better at pod scale; the crossing is the result.

	T (lower is better)		merge depth	
Placement	Boardfly	Shiftfly	Boardfly	Shiftfly
random	2.768	2.470	2.868	2.772
locality-aware	2.174	**2.112**	2.336	**2.152**
saving from placement	−21.5%	−14.5%		

Locality-aware placement removes 21.5% of tree cost on Boardfly and 14.5% on Shiftfly. **Shiftfly's residual advantage at matched placement is** 2.9%, with merge depth 7.9% lower. The overwhelming majority of the available saving on shared content therefore comes from *placement*, which is topology-agnostic, and not from the shift algebra.

Furthermore the efficiency ratio η ranks Boardfly *higher* (1.431 against 1.364) while absolute tree cost ranks it *lower* (2.174 against 2.112)—precisely the inversion predicted in Remark 7. Had we reported η alone, as the natural reading of the objective in (2) invites, we would have drawn the opposite conclusion.

Conclusion of the evaluation. The defensible case for Shiftfly is scaling and expansion. The redundancy case, as measured here, is *primarily an argument for locality-aware placement*, which is topology-agnostic.

8 Limitations, and how this could be wrong

1. **Boardfly is better at the scale it was designed for.** At 1,152 chips the complete global tier is simply the right answer. Nothing here contradicts that choice.
2. **Bisection.** De Bruijn and Kautz families have bisection width $\Theta(N/\log N)$, against $\Theta(N)$ for a complete or expander tier. Measured against a *hierarchy* the comparison favours Shiftfly, because hierarchies expand badly; measured against a flat high-radix fabric it would not. All-to-all-dominated workloads are the adversarial case.

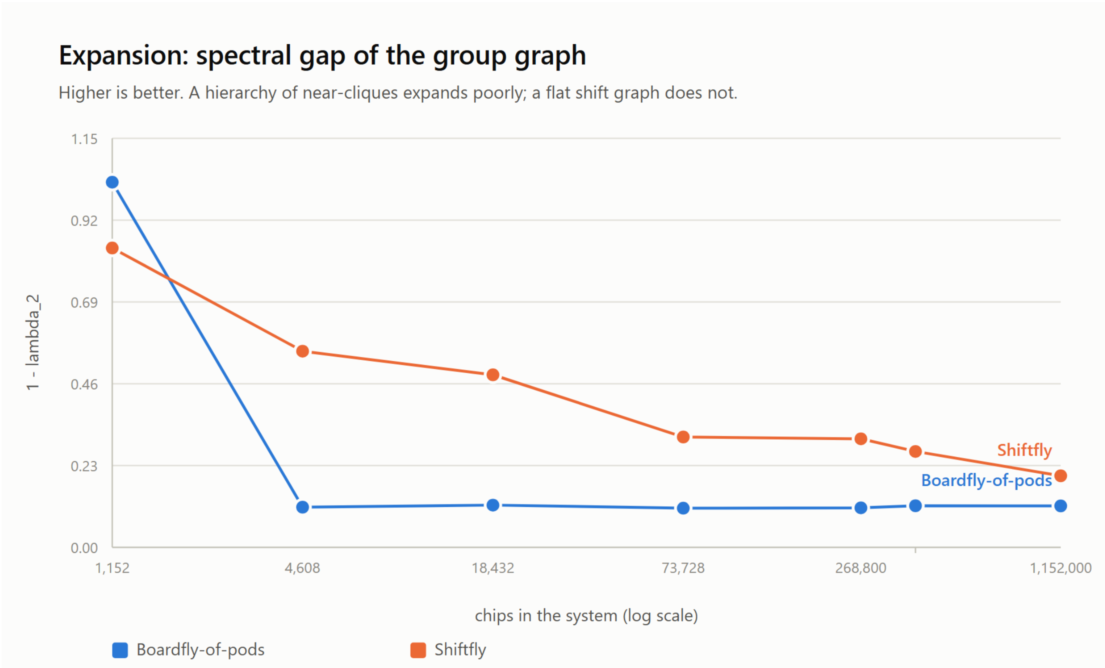

Figure 2: Spectral gap of the group graph. A hierarchy of near-cliques joined by few links expands poorly whatever its diameter; the flat shift graph does not.

3. **Cabling.** Shift edges are long and irregular, which is the standard reason de Bruijn networks are not built. The mitigation is specific and not general: the global tier is already optically circuit-switched, so the wiring is a permutation the switch installs, not copper anyone routes. This argument is available to an operator that owns an OCS layer and to essentially nobody else.

4. **Deterministic routing.** Shift routing yields one path. Load balance will require non-minimal alternatives, and randomising the route partly destroys the merge property of Theorem 13. The interaction is unquantified here.

5. **The switch is mechanical.** MEMS-based optical circuit switches reconfigure in milliseconds to seconds, so the Kautz permutation is fixed at scheduling time. Label assignment is therefore a scheduling decision, not a runtime one.

6. **The workload model is synthetic.** Cluster count, agent fan-out, scheduler locality and Zipf exponent are chosen to be plausible, not measured. The sharing conclusions are only as good as those four numbers.

9 Related work

Dragonfly established the high-radix hierarchical form that Boardfly follows. The degree–diameter literature supplies the ceiling used in Section 3, and recent topologies—Slim Fly from McKay–Miller–Širáň graphs, PolarFly from Erdős–Rényi polarity graphs, Bundlefly—pursue it directly at router radix. Shiftfly differs in operating at accelerator radix, where the endpoint *is* the router, and in taking the content-addressing property of the shift families seriously; the same algebra underlies the Koorde distributed hash table. Kautz digraphs are due to Kautz; the arbitrary-order generalizations to Reddy, Pradhan and Kuhl, and to Imase and Itoh.

10 Conclusion

Google's move from torus to Boardfly traded a $\Theta(N^{1/3})$ diameter for a constant one and was clearly correct for a 1,152-chip inference pod. The mechanism it used—full connectivity at every tier—does not extend, because complete graphs need $N-1$ ports. Replacing only the global tier with a generalized Kautz digraph keeps the diameter logarithmic in the number of groups at a fixed port budget, yields table-free routing and native content addressing, and measurably improves expansion.

The redundancy argument that motivated the design did not survive its own evaluation intact: most of the available saving on shared content comes from placement, not from topology. We regard reporting that as more useful than the alternative.

A Reproduction

```
git clone https://github.com/EylonKrause/shiftfly
cd shiftfly
python3 run_experiments.py           # stdlib only
python3 -m unittest discover -s tests
```